

BALLENAS ISLAND, BC, Canada

WMO: 717690

Lat: **49.3506N**Lon: **124.1606W**Elev: **33**StdP: **14.68**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.7	33.3	19.0	14.3	33.4	23.1	17.4	36.1	37.8	40.0	35.3	42.7	10.5	230	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	74.3	65.9	72.2	65.0	70.4	63.9	67.4	72.4	65.9	70.7	64.6	69.4	10.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.0	70.8	63.8	88.7	69.2	62.4	84.5	67.9	31.9	72.7	30.7	70.9	29.7	69.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.5	30.7	27.1	28.0	79.5	2.9	3.1	25.8	81.7	24.1	83.5	22.5	85.3	20.3	87.6		
(5)			24.7	68.8	2.5	2.1	22.9	70.3	21.4	71.5	20.0	72.7	18.2	74.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	42.7	42.5	45.4	49.4	56.0	60.6	65.2	65.2	59.8	52.3	46.0	42.7	
	DBStd	9.28	4.76	4.10	3.59	3.63	4.26	4.18	3.49	3.41	3.60	3.50	4.49	4.23	
	HDD50	1015	229	209	148	52	4	0	0	0	0	16	130	227	
	HDD65	4719	693	629	609	467	280	146	40	40	161	394	569	691	
	CDD50	1879	1	0	4	34	189	318	470	470	294	87	11	1	
	CDD65	109	0	0	0	0	1	14	45	45	4	0	0	0	
	CDH74	79	0	0	0	0	0	14	38	27	0	0	0	0	
	CDH80	3	0	0	0	0	0	1	1	1	0	0	0	0	
Wind	WSAvg	12.5	13.7	12.4	12.7	11.9	11.2	11.3	11.9	11.2	11.3	12.9	14.2	14.8	
Precipitation	PrecAvg	34.9	5.1	3.0	3.2	2.0	1.7	1.7	0.9	1.0	1.6	4.0	5.0	5.7	
	PrecMax	42.9	9.9	6.1	6.5	4.0	3.5	3.1	1.8	3.8	4.5	8.8	11.4	8.1	
	PrecMin	23.3	2.2	0.1	0.7	0.8	0.4	0.5	0.2	0.1	0.2	1.3	1.5	2.0	
	PrecStd	5.7	2.0	1.4	1.4	0.9	0.7	0.8	0.5	0.8	1.1	1.9	2.5	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	51.7	55.5	62.8	69.9	75.8	77.9	77.0	71.6	63.5	57.3	53.9	
		MCWB	49.2	48.0	50.4	55.5	59.9	64.4	68.5	67.1	63.8	60.3	54.6	50.8	
	2%	DB	50.8	49.6	53.0	59.1	66.3	72.0	74.5	74.1	68.6	60.1	54.4	51.0	
		MCWB	48.0	46.7	49.2	53.7	58.8	63.7	66.4	66.1	63.0	57.6	52.3	48.6	
	5%	DB	49.3	48.4	51.3	56.4	64.1	69.0	72.3	71.9	66.5	58.4	53.0	49.5	
		MCWB	47.2	46.0	47.9	51.8	57.8	62.1	65.2	65.3	61.8	55.9	51.4	47.5	
	10%	DB	48.2	47.4	49.9	54.5	61.9	66.4	70.3	70.1	64.8	56.8	51.7	48.2	
		MCWB	46.6	45.3	47.0	50.3	56.6	60.4	64.0	64.3	60.6	54.4	50.2	46.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	49.3	52.1	56.8	61.8	67.0	69.9	70.3	65.6	61.4	55.4	52.1
			MCDB	51.7	50.2	54.4	61.2	67.0	72.9	76.2	74.4	69.4	62.8	56.8	53.0
2%		WB	49.1	48.0	50.1	54.4	59.8	64.4	67.7	67.5	63.9	58.5	53.2	49.5	
		MCDB	49.9	48.9	52.0	58.1	64.4	70.1	73.1	72.0	67.5	59.7	53.9	50.5	
5%		WB	47.9	46.8	48.6	52.6	58.5	62.5	66.1	66.0	62.5	56.8	51.9	48.1	
		MCDB	48.8	47.8	50.4	55.7	62.8	67.7	71.3	70.5	65.8	57.9	52.8	49.1	
10%		WB	46.8	45.8	47.4	50.9	57.3	60.8	64.7	64.8	61.2	54.9	50.4	46.7	
		MCDB	47.7	47.0	49.2	53.5	61.5	65.5	69.6	69.3	64.3	56.2	51.5	47.9	
Mean Daily Temperature Range		5% DB	MDBR	4.6	5.2	6.2	7.1	7.9	8.1	8.3	7.7	6.6	5.4	5.3	4.6
			MCDBR	6.0	5.7	7.7	9.5	10.5	10.8	10.6	9.7	8.3	7.3	6.9	6.1
	MCWBR		4.8	4.6	5.6	6.5	7.0	7.0	6.9	6.6	6.0	6.1	6.2	5.0	
	5% WB	MCDBR	5.6	5.3	7.1	9.0	9.5	9.8	9.9	8.9	7.8	6.8	6.7	6.1	
		MCWBR	4.8	4.7	5.6	6.5	6.7	7.0	6.9	6.6	6.1	6.1	6.2	5.4	
Clear-Sky Solar Irradiance	taub	0.315	0.307	0.328	0.346	0.367	0.360	0.348	0.350	0.334	0.332	0.314	0.308		
	taud	2.499	2.568	2.520	2.475	2.434	2.469	2.548	2.548	2.582	2.559	2.546	2.486		
	Ebn at Noon	237	272	282	287	283	285	287	282	277	257	237	223		
	Edh at Noon	19	22	28	32	35	34	31	29	26	22	18	17		
All-Sky Solar Radiation	RadAvg	267	547	814	1268	1649	1776	1955	1666	1172	637	317	210		
	RadStd	40	85	109	128	162	202	184	138	102	97	58	32		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	-1.03	-2.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page